
Week # 2

Course: Data Science

Class: MSCS

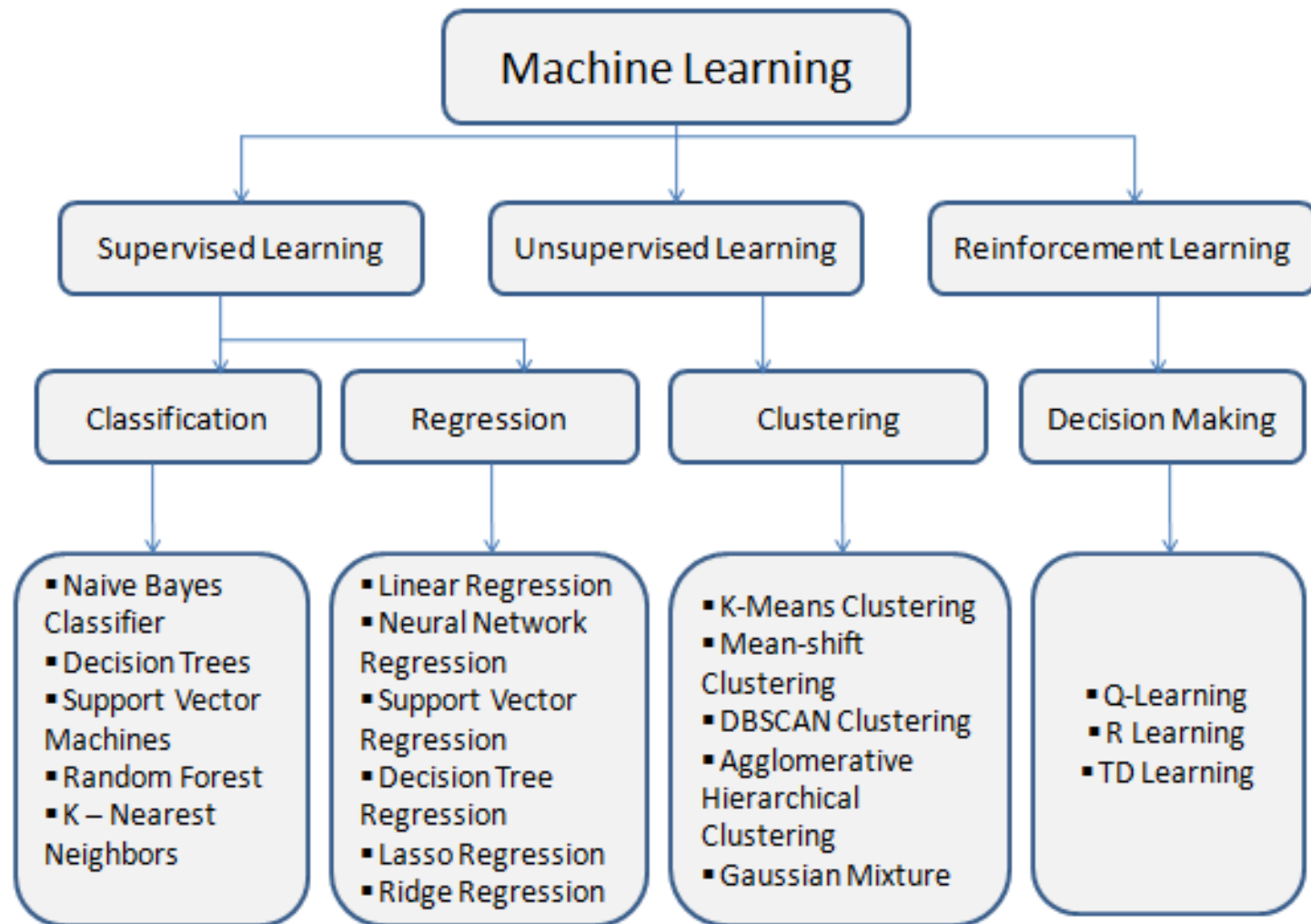
Instructor: Abdul Rehman



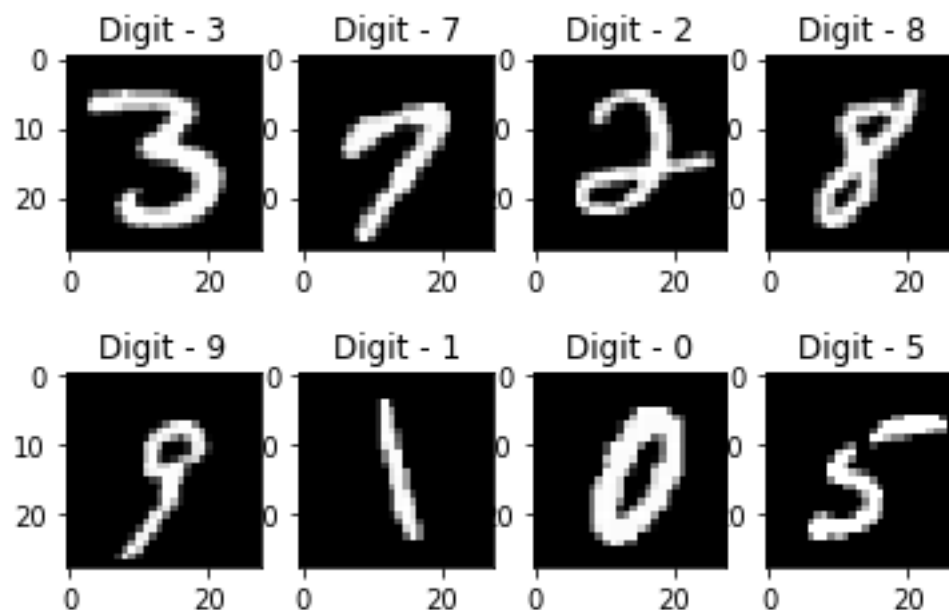
Lecture Outlines

- ❑ Types of Machine Learning:
 1. Supervised Learning
 2. Unsupervised Learning
 3. Semi-Supervised Learning
 4. Re-inforcement Learning
- ❑ Important Concepts related to ML/DS
- ❑ Basic Steps of ML and Data Science

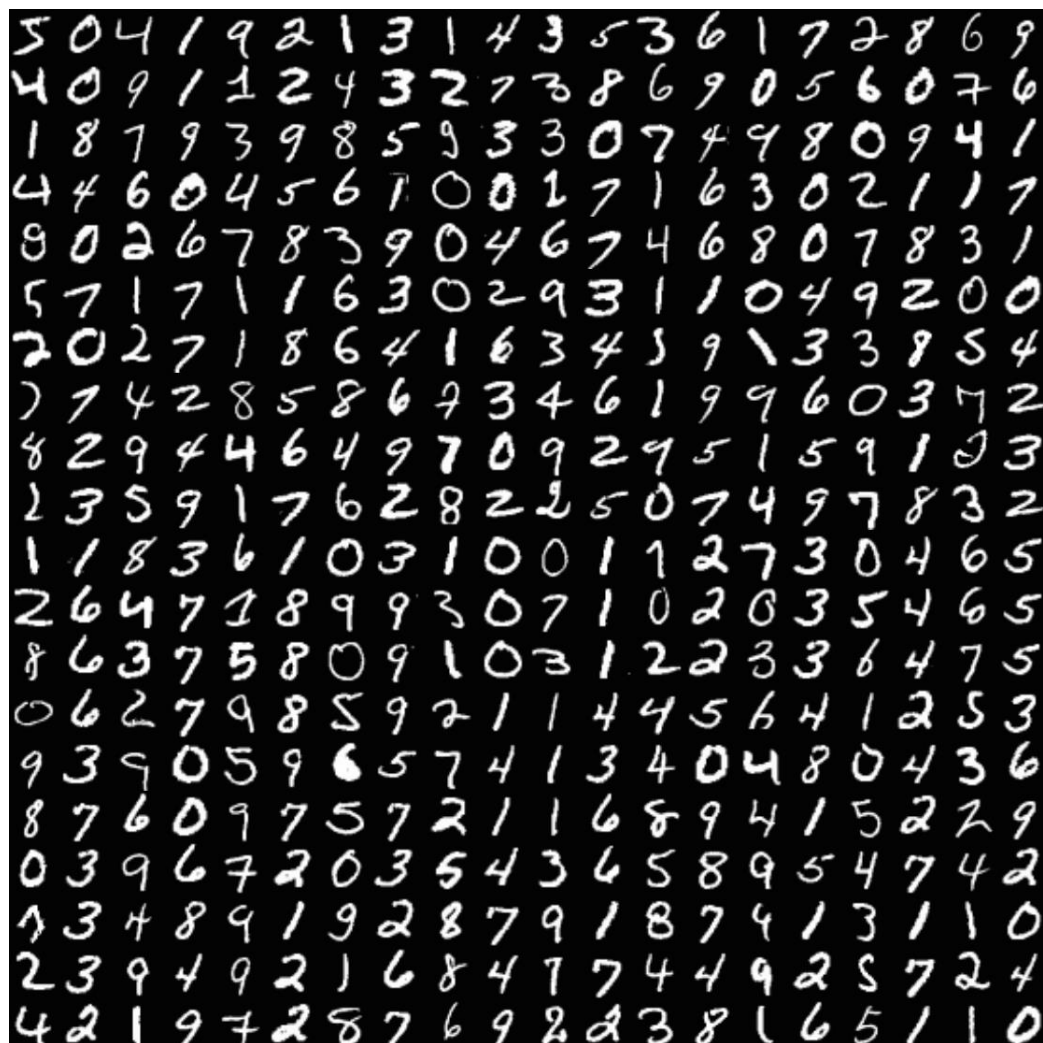




Handwritten Digits Supervised ML



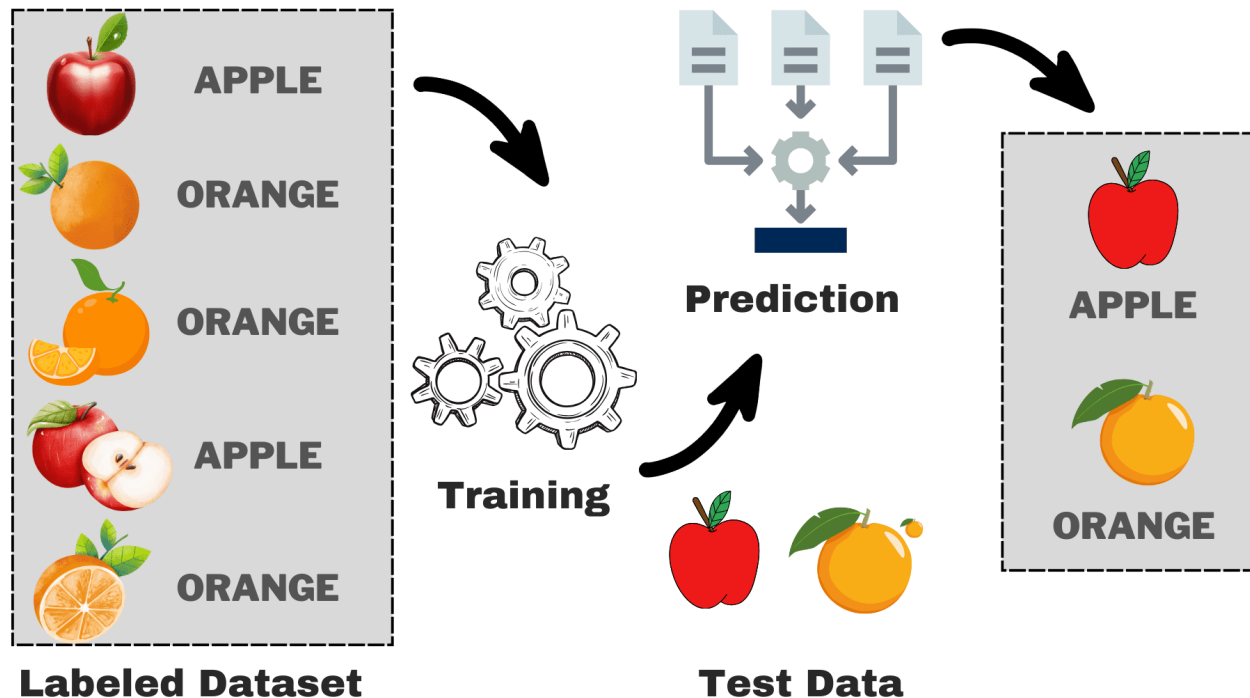
Handwritten Digits Un-Supervised ML





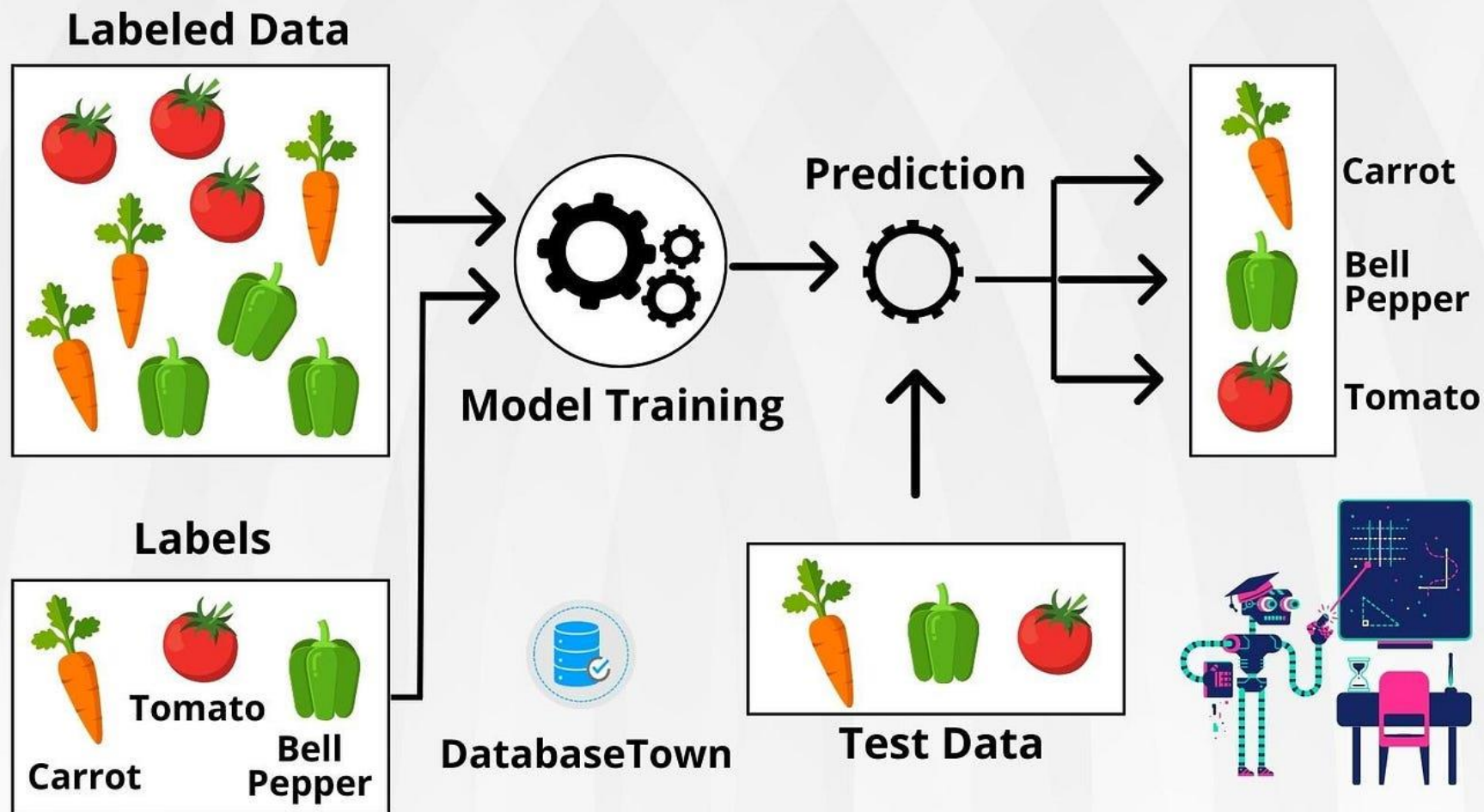
1. Supervised Learning

- Supervised learning involves **training a model on labeled data**
- The given data is labeled. Both **classification** and **regression** problems are supervised learning problems.



SUPERVISED LEARNING

Supervised machine learning is a branch of artificial intelligence that focuses on training models to make predictions or decisions based on labeled training data.



Examples of Supervised ML

1. Email Spam Detection:

The system is trained with emails labeled as “*spam*” or “*not spam*.”

After learning, it can classify new emails correctly.

2. Predicting House Prices:

Given data such as **size**, **location**, and number of **rooms** (*inputs*) and known house prices (*output*),

the model learns to predict prices for new houses.



Supervised Learning Classifiers

- ☐ Linear Regression
- ☐ Logistic Regression
- ☐ Decision Trees
- ☐ Random Forest
- ☐ Naïve Bays
- ☐ AdaBoost
- ☐ Gradient Boosting
- ☐ KNN
- ☐ Support Vector Machines (SVM)
- ☐ Neural Networks



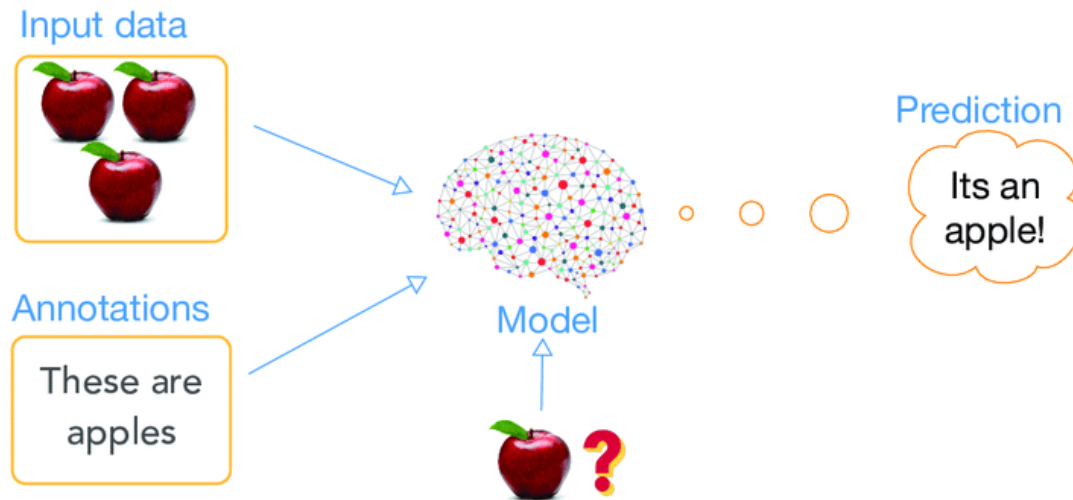
2. Un-Supervised Learning

- In unsupervised learning, the data is **unlabeled** — the model is not told what the correct output is.
- It must **find hidden patterns, structures, or relationships** in the data on its own.

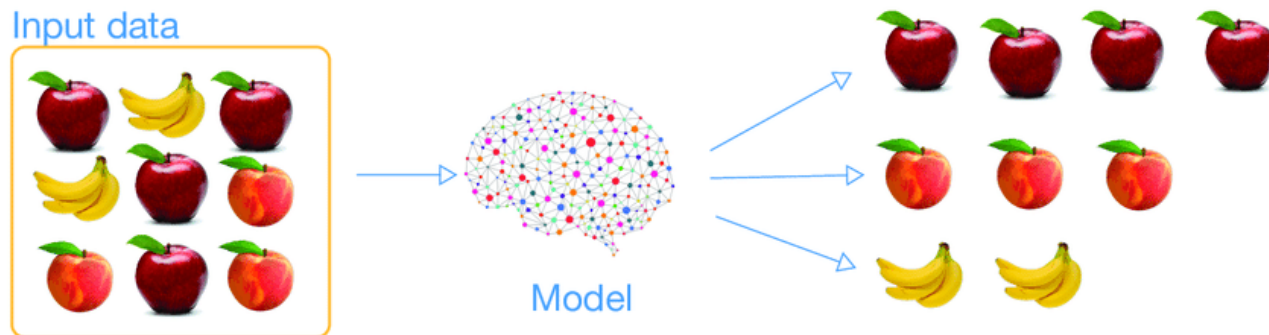


Examples

supervised learning



unsupervised learning



Common Classifiers of unsupervised ML

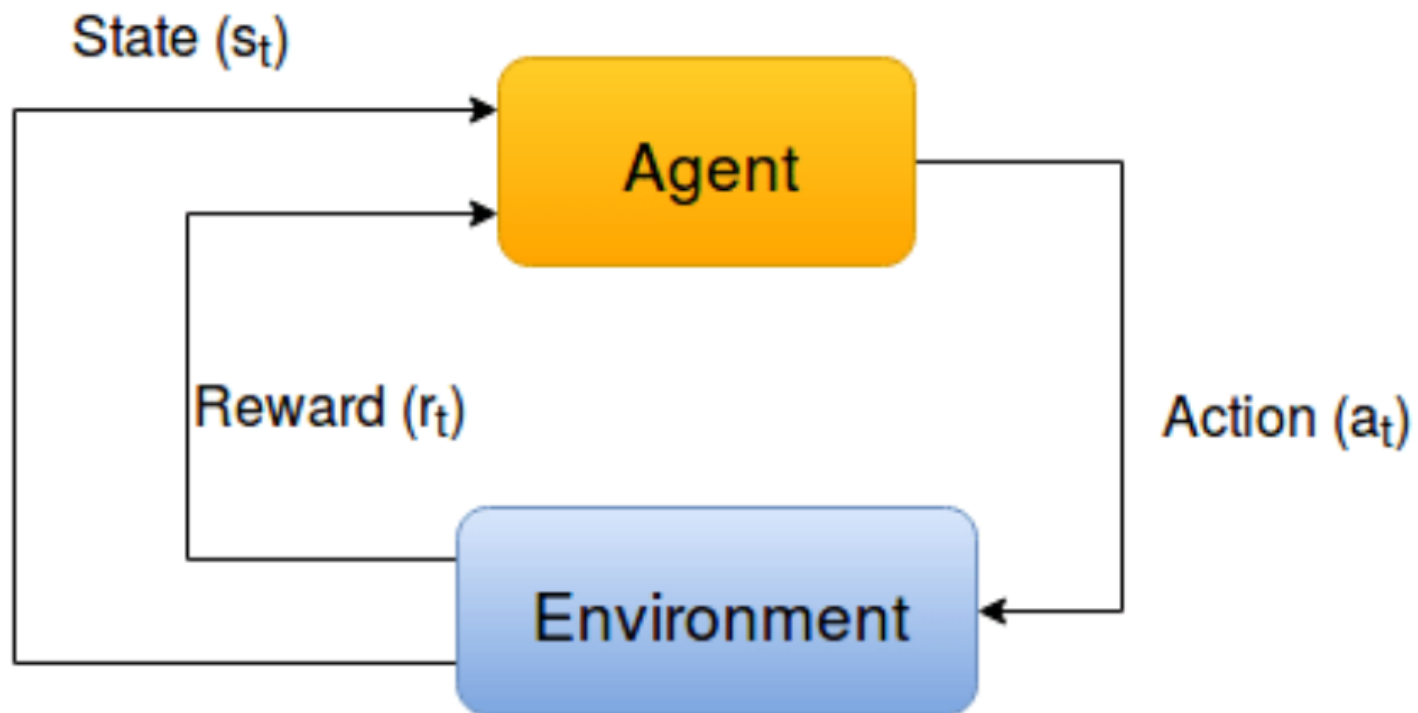
- ☐ K-Means Clustering
- ☐ DBSCAN Clustering Algorithm
- ☐ Hierarchical Clustering
- ☐ Principal Component Analysis (PCA)
- ☐ Apriori Algorithm



3. Reinforcement learning

- Reinforcement learning involves **training a model through trial and error.**
- An agent interacts with an environment, takes actions, and learns from **rewards or penalties.**
- Reinforcement learning is the problem of getting an agent to act in the world so as to maximize its rewards.





Example of Reinforcement learning

- **Example:**
- Imagine you're in class. If you pay attention and do your homework, your teacher gives you a gold star. This makes you happy, so you keep paying attention and doing your homework.
- But if you talk during the class instead of listening, your teacher gives you a warning. This makes you feel bad, so you try not to talk during class anymore.
- You learn that paying attention and doing homework gets you rewards, while talking during class gets you punishments. This is called "**reinforcement learning**."

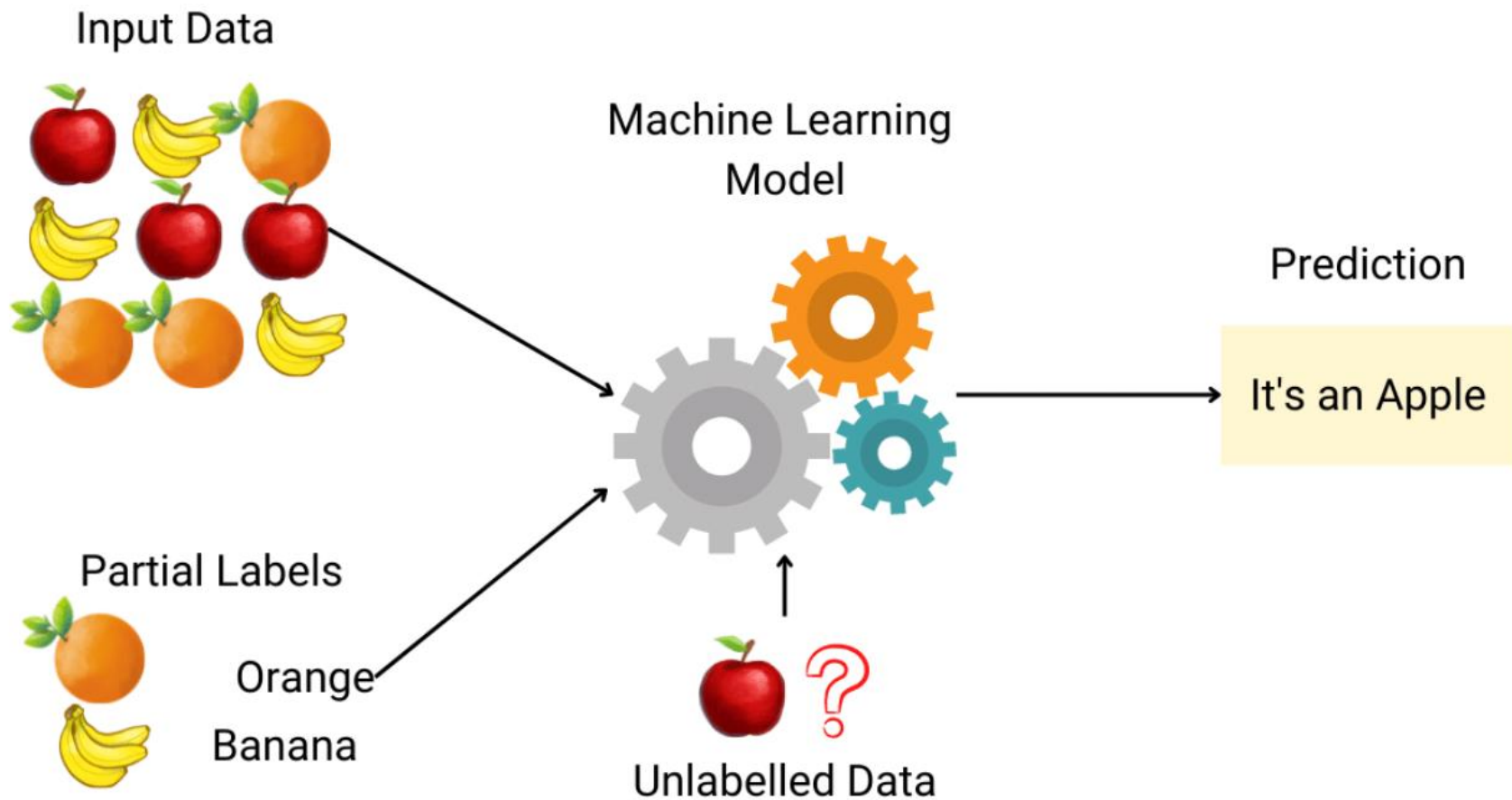


4. Semi-Supervised Learning

- Problems where you have a large amount of input data and only some of the data is labeled, are called semi-supervised learning problems.
- For example, a photo archive where only some of the images are labeled, (e.g. dog, cat, person) and the majority are unlabeled
- Semi-supervised learning is an approach to machine learning that combines small labeled data with a large amount of unlabeled data during training.



4. Semi-Supervised Learning



Applications of Machine Learning

Machine learning is used in a wide variety of applications, including

1. **Image classification:** Identify objects, faces, and other features in images.
2. **Speech recognition:** Convert spoken language into text.
3. **Recommendation systems:** Make personalized recommendations to users.
4. **Predictive analytics:** Predict outcomes, such as sales, customer churn, and stock prices.
5. **Medical diagnosis:** Detect diseases and other medical conditions.
6. **Fraud detection:** Identify fraudulent transactions.
7. **Email spam detection:** Classify emails as spam or not spam.
8. **Autonomous vehicles**



Applications of Machine Learning

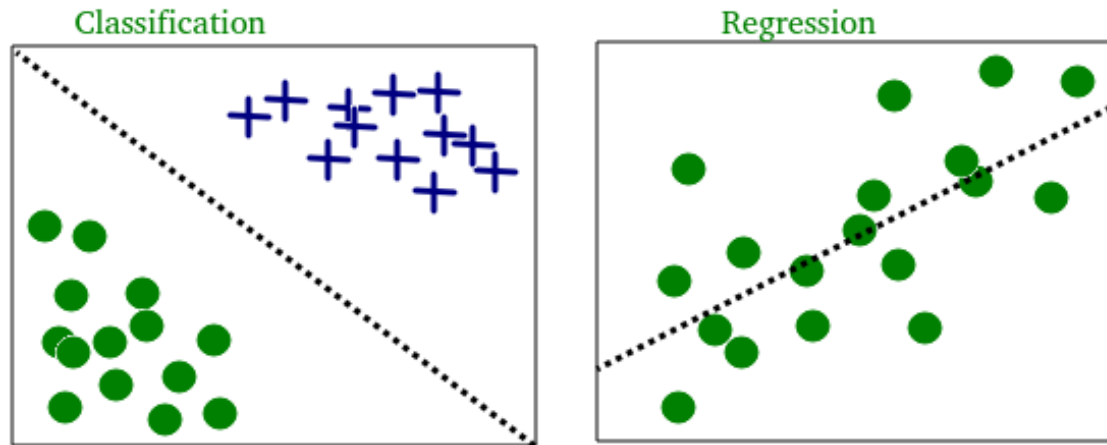
9. **Sports analytics:** Analyze player performance, make game predictions, and optimize strategies.
10. **Content recommendation:** Classify and tag content to make it easier to recommend similar items to users.
11. **Market basket analysis:** Discover associations between products.
12. **Genomic data analysis:** Identify patterns or group genes with similar expression profiles.
13. **Customer behavior analysis:** Uncover patterns and insights for better marketing and product recommendations.
14. **Gaming:** Recognize characters, analyze player behavior.
15. **Weather forecasting:** Make predictions for temperature, precipitation, and other meteorological parameters.



Classification, Regression, and Clustering

1. Classification:

- It is a supervised learning technique, and it predicts
 - a discrete label or category for given input data.
- **Examples** include spam detection, and image recognition.
- **Models:** Logistic Regression, (SVM), (K-NN), Decision Trees, Random Forests, Neural Networks, Naive Bayes.



Classification, Regression, and Clustering

2. Regression:

- **It is a** Supervised Machine Learning technique.
- It predict a **continuous value** based on input features.
- **Examples** include predicting
- **house prices, weather forecasting, sales prediction, and medical diagnosis, etc.**
- **Models:**
- Linear Regression, Polynomial Regression, Support Vector Regression (SVR), Decision Trees, Random Forests, Neural Networks.



Classification, Regression, and Clustering

3. Clustering:

- **Clustering is Unsupervised** Machine Learning technique
- Clustering tasks involve **grouping data points into clusters based on similarity without using labeled responses.**
- **Examples:**
- **Sentiment Analysis:** Categorizing customer reviews as "positive," "negative," or "neutral."
- **Medical Diagnosis (Cancer Detection):** Predicting whether a tumor is benign or malignant.
- **Models:** K-Means Clustering, Hierarchical Clustering, DBSCAN, etc.



Important Concepts related to ML/DS/AI

1. Model
2. Feature
3. Target variable, label
4. Feature
5. Training
6. Testing



Basic Steps of ML and Data Science

1. Problem Definition

- Define the problem you want to solve.
- Identify the type of machine learning problem (regression, classification, clustering, etc.).

2. Data Collection

- Gather relevant data from databases, APIs, web scraping, surveys, or sensors.
- Ensure sufficient quality and quantity of data



Basic Steps of ML and Data Science Continue...

3. Data Preprocessing & Cleaning

- Handle missing values.
- Remove duplicates and irrelevant features.
- Normalize or standardize numerical data.

4. Exploratory Data Analysis (EDA)

- Visualize data using graphs and plots.
- Identify patterns, correlations, and outliers.



Basic Steps of ML and Data Science Continue...

5. Feature Engineering

- Select or create relevant features to improve model performance.
- Use techniques like one-hot encoding, feature scaling, and dimensionality reduction.

6. Model Selection

- Choose the right algorithm (e.g., Decision Trees, SVM, Neural Networks).
- Split data into training, validation, and test sets.



Basic Steps of ML and Data Science Continue...

7. Model Training

- Train the model using the training data.
- Optimize parameters using techniques like grid search or random search.

8. Model Evaluation

- Measure performance using metrics like accuracy, precision, recall, MAE, RMSE, or F1-score.
- Compare different models and tune hyperparameters.



Basic Steps of ML and Data Science Continue...

9. Model Deployment

- Deploy the model into a real-world system (e.g., web apps, APIs, mobile apps).
- Monitor its performance in production.

10. Model Maintenance & Improvement

- Continuously update the model with new data.
- Retrain and fine-tune it for better accuracy.

